

SO SIMPLE, YET SO EFFECTIVE

Ottavia M. Epifania
University of Trento
`ottavia.epifania@unitn.it`

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How it started:

A random lesson by Professor Pastore on Generalized Linear Mixed Effects Models

How it started:

A random lesson by Professor Pastore on Generalized Linear Mixed Effects Models

How it ended:



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Psychological Methods

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A Guided Tutorial on Linear Mixed-Effects Models for the Analysis of Accuracies and Response Times in Experiments With Fully Crossed Design

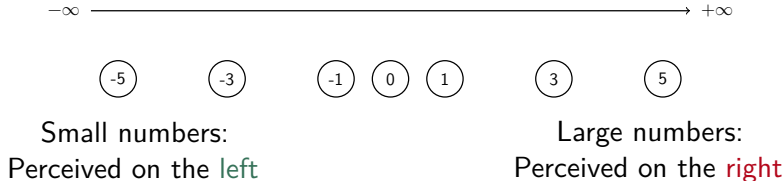
Ottavia M. Epifania, Pasquale Anselmi, and Egidio Robusto

Department of Philosophy, Sociology, Education and Applied Psychology, University of Padova

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FULLY-CROSSED STRUCTURES

AN EXAMPLE: THE SNARC EFFECT



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A *sample* of small numbers:

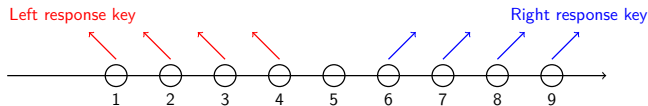
1, 2, 3, 4

A *sample* of large numbers:

6, 7, 8, 9

Two conditions:

The “natural” one (so-called *compatible* condition)



AN EXAMPLE: THE SNARC EFFECT

A *sample* of small numbers:

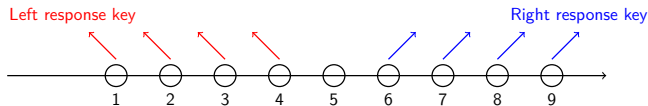
1, 2, 3, 4

A *sample* of large numbers:

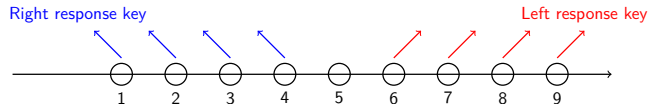
6, 7, 8, 9

Two conditions:

The “natural” one (so-called *compatible* condition)



The “innatural” one (so-called *incompatible* condition)



AN EXAMPLE: THE SNARC EFFECT

$t = \{1, 2, \dots, T\}$: Number of trials (condition $\times$ stimulus $\times$ respondent)

		Small Numbers				Large Numbers			
	Condition	1	2	3	4	6	7	8	9
Jane	Compatible	y_{cj1}	y_{cj2}	y_{cj3}	y_{cj4}	y_{cj6}	y_{cj7}	y_{cj8}	$\sum_{t=1}^T y_{cj}/T$
	Incompatible	y_{ij1}	y_{ij2}	y_{ij3}	y_{ij4}	y_{ij6}	y_{ij7}	y_{ij8}	$\sum_{t=1}^T y_{ij}/T$
Mario	Compatible	y_{cm1}	y_{cm2}	y_{cm3}	y_{cm4}	y_{cm6}	y_{cm7}	y_{cm8}	$\sum_{t=1}^T y_{cm}/T$
	Incompatible	y_{im1}	y_{im2}	y_{im3}	y_{im4}	y_{im6}	y_{im7}	y_{im8}	$\sum_{t=1}^T y_{im}/T$

SCORING

Person-level scores

$$s_p = \frac{\bar{X}_{p,\text{comp}} - \bar{X}_{p,\text{inc}}}{sd_{\text{pooled}}}$$

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Advantages

Ease of computation

Ease of interpretation

Person-level scores

$$s_p = \frac{\bar{X}_{p,\text{comp}} - \bar{X}_{p,\text{inc}}}{sd_{\text{pooled}}}$$



Advantages

Ease of computation

Ease of interpretation



(Implicit) Assumptions

- ① Being slow (less accurate) in one condition = being fast (or more accurate) in the opposite one: 0 means absence of bias
- ② All stimuli have the same impact (fixed effects)

A long tradition

i Respondents are random factors

Sampled from a larger population

Need for acknowledging the sampling variability

Results can be generalized to other respondents belonging to the same population

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i Respondents are random factors

Sampled from a larger population

Need for acknowledging the sampling variability

Results can be generalized to other respondents belonging to the same population

i Stimuli/items are fixed factors

Taken to be entire population

There is no sampling variability

There is no need to generalize the results because the stimuli are the population

With long lasting consequences

- Generalization of the results is impaired
- Error variance everywhere, left free to bias everything
- The information at the stimulus level is lost

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$$\Sigma$$

Linear Mixed Effects Models

$$\psi$$

Rasch model

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 Σ

Linear Mixed Effects Models

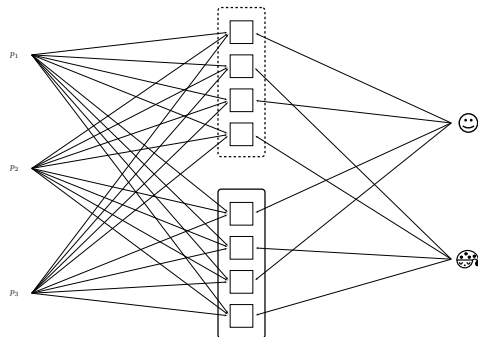
 ψ

Rasch model



Rasch-like parametrization estimated with Linear Mixed Effects Models

WHEN?



Sample-level differences:

Compatible and incompatible can be defined *a priori* (SNARC effect)

Individual differences:

Compatible and incompatible are defined within each respondent (Implicit Association Test)

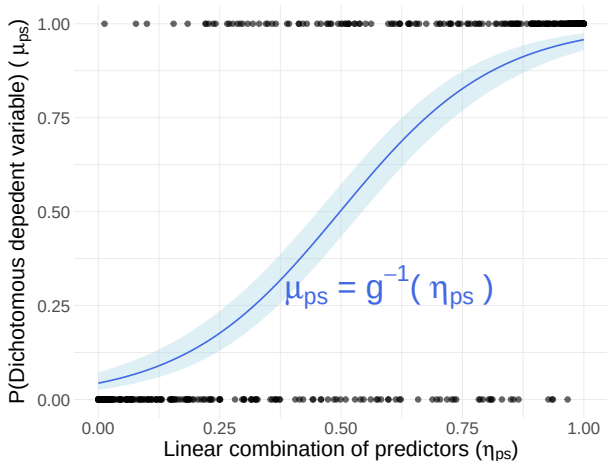
A CLASSIC OF PSYCHOMETRICS

The Rasch Model

$$P(x_{ps} = 1 | \theta_p, b_s) = \frac{\exp(\theta_p - b_s)}{1 + \exp(\theta_p - b_s)}$$

θ_p : Latent trait of person p

b_s : “challenging” power of stimulus s



Logit link function g :

$$g(\eta_{ps}) = \log \left(\frac{\mu_{ps}}{1 - \mu_{ps}} \right)$$

Inverse g^{-1}

$$g^{-1} = \frac{\exp(\eta_{ps})}{1 + \exp(\eta_{ps})}$$

The log-normal model

$$E(t_{ps}|\tau_p, \delta_s) = \delta_s - \tau_p + \varepsilon$$

τ_p : the speed of person p

δ_s : the time intensity of stimulus s

The log-normal model

$$E(t_{ps}|\tau_p, \delta_s) = \delta_s - \tau_p + \varepsilon$$

τ_p : the speed of person p

δ_s : the time intensity of stimulus s

A linear model with an identity function!

i Rasch

$$P(x_{ps} = 1) = \frac{\exp(\theta_p - b_z)}{1 + \exp(\theta_p - b_z)}$$

i Log-normal

$$E(t_{ps} | \tau_p, \delta_s) = \delta_s - \tau_p + \varepsilon$$

! GLM (inverse function)

$$P(x_{ps} = 1) = \frac{\exp(\theta_p + b_s)}{1 + \exp(\theta_p + b_s)}$$

! LM (identity function)

$$E(t_{ps} | \tau_p, \delta_s) = \delta_s + \tau_p + \varepsilon$$

In a LM:

$$\eta = \mathbf{X}\beta$$

$\mathbf{X}$: Model Matrix

β : Coefficients

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$$\eta = \mathbf{X}\beta$$

$\mathbf{X}$: Model Matrix

β : Coefficients

Needs to be extended:

$$\eta = \mathbf{X}\beta + \mathbf{Z}d$$

d : Random effects associated to the random factors in Z ... Not model parameters! *Best Linear Unbiased Predictors*

Γ : Parameters estimated for the random factors in the model (variances and covariances)

The maximal model

Address all the possible sources of random variability that can be expected

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The models that are useful for ones aim

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Common goal: Investigate the changes in the performance of the respondents between the associative conditions

The maximal model

Address all the possible sources of random variability that can be expected

The models that are useful for ones aim

Common goal: Investigate the changes in the performance of the respondents between the associative conditions

Less common: Investigate the changes in the functioning of the stimuli between the associative conditions

Preliminaries

Index	Meaning	Variable
$p = 1, \dots, P$	Respondent	respondents
$s = 1, \dots, S$	Stimulus	stimuli
$c \in \{0, 1\}$	Associative condition	condition
i	Trial	

Accuracy:

GLMM

$$y = [0, 1]$$

Log-time response

LMM

$$y = [0, +\infty) \text{ (log-transformed)}$$

$$\varepsilon \sim \mathcal{N}(0, \sigma^2)$$

Model 1

 Mathematical Notation

$$y = \beta_c X_c + \alpha_p[i] + \alpha_s[i]$$

 lme4 notation

```
y ~ 0 + condition + (1|stimuli) + (1|respondents)
```

 Rasch-like parametrization

	GLMM	LMM
respondents	θ_p	τ_p
stimuli	b_s	δ_s

Model 2

 Mathematical Notation

$$y = \beta_c X_c + \alpha_p[i] + \beta_s[i] c_i$$

 lme4 notation

```
y ~ 0 + condition + (0+condition|stimuli) + (1|respondents)
```

 Rasch-like parametrization

	GLMM	LMM
respondents	θ_p	τ_p
stimuli	b_{sc}	δ_{sc}

Model 3

 Mathematical Notation

$$y = \beta_c X_c + \beta_p[i]c_i + \alpha_s[i]$$

 lme4 notation

```
y ~ 0 + condition + (1|stimuli) + (0+condition|respondents)
```

 Rasch-like parametrization

	GLMM	LMM
respondents	θ_{pc}	τ_{pc}
stimuli	b_s	δ_s

All models are wrong...

Find the useful model via model comparison: AIC and BIC

The lower the value, the better the model

! AIC, BIC, and model complexity:

Total number of parameters: β and Γ
NOT the levels in d

Model 2 and Model 3: Same complexity, different focus

The chosen model is the least wrong model *given the models considered*: Relativity applies everywhere

Coke
Good

Pepsi
Bad

Check the categories – Press Space Bar to continue

Coke
Good



Pepsi
Bad

Response key: E

Coke
Good



Pepsi
Bad

Response key: I

Coke
Good

Pepsi
Bad

terrible

Response key: I

Coke
Good

Pepsi
Bad

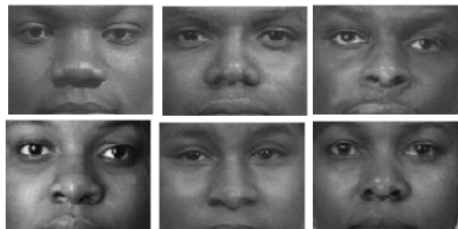
glory

12 OBJECT STIMULI

White people faces



Black people faces



16 ATTRIBUTE STIMULI

Positive attributes

Good, laughter, pleasure, glory, peace, happy,
joy, love

Negative attributes

Evil, bad, horrible, terrible, nasty, pain,
failure, hate

Get set

```
# install package for fitting lmms
install.packages("lme4")
# nice plots :) 
install.packages("ggplot2")

library(lme4)
library(ggplot2)
```

The data

```
data = read.csv("data/example-data.csv",  
                header = TRUE, sep = ",")  
head(data)
```

	respondent	condition	stimuli	accuracy	latency
1	1	Whitegood	hate	1	1224
2	1	Whitegood	bf14	1	5160
3	1	Whitegood	laughter	1	1214
4	1	Whitegood	bf56	1	1143
5	1	Whitegood	evil	1	827
6	1	Whitegood	wf3	1	1859

Number of trials $\times$ condition $\times$ respondent:

```
table(data$respondent, data$condition)
```

	Whitebad	Whitegood
1	60	60
2	60	60
3	60	60
4	60	60
5	60	60
6	60	60
...		

GLMMs FOR ACCURACY

$$\text{Model 1: } y_i = \text{logit}^{-1}(\alpha + \beta_c X_c + \alpha_{p[i]} + \alpha_{s[i]})$$

```
accuracy1 = glmer(accuracy ~ 0 + condition + (1|stimuli) + (1|respondent),  
  data = data,  
  family = "binomial")
```

$$\text{Model 2: } y_i = \text{logit}^{-1}(\alpha + \beta_c X_c + \alpha_{p[i]} + \beta_{s[i]} c_i)$$

```
accuracy2 = glmer(accuracy ~ 0 + condition + (0 + condition|stimuli) +  
  (1|respondent),  
  data = data,  
  family = "binomial")
```

$$\text{Model 3: } y_i = \text{logit}^{-1}(\alpha + \beta_c X_c + \alpha_{s[i]} + \beta_{p[i]} c_i)$$

```
accuracy3 = glmer(accuracy ~ 0 + condition + (1|stimuli) +  
  (0 + condition|respondent),  
  data = data,  
  family = "binomial")
```

LMMs FOR LOG-TIME RESPONSES

$$\text{Model 1: } y_i = \alpha + \beta_c X_c + \alpha_{p[i]} + \alpha_{s[i]} + \varepsilon_i$$

```
logtime1 = lmer(log(latency) ~ 0 + condition + (1|stimuli) + (1|respondent),  
               data = data,  
               REML = FALSE)
```

$$\text{Model 2: } y_i = \alpha + \beta_c X_c + \alpha_{p[i]} + \beta_{s[i]} c_i + \varepsilon_i$$

```
logtime2 = lmer(log(latency) ~ 0 + condition + (0 + condition|stimuli) +  
               (1|respondent),  
               data = data,  
               REML = FALSE)
```

$$\text{Model 3: } y_i = \alpha + \beta_c X_c + \alpha_{s[i]} + \beta_{p[i]} c_i + \varepsilon_i$$

```
logtime3 = lmer(log(latency) ~ 0 + condition + (1|stimuli) +  
               (0 + condition|respondent),  
               data = data,  
               REML = FALSE)
```

GLMMs

 Important!

The use of the `anova()` function is just for the convenience of having all the information on the same page!

```
anova(accuracy1, accuracy2, accuracy3)
```

Data: data

Models:

```
accuracy1: accuracy ~ 0 + condition + (1 | stimuli) + (1 | respondent)
```

```
accuracy2: accuracy ~ 0 + condition + (0 + condition | stimuli) + (1 | respondent)
```

```
accuracy3: accuracy ~ 0 + condition + (1 | stimuli) + (0 + condition | respondent)
```

	npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
accuracy1	4	4144.3	4172.1	-2068.1	4136.3			
accuracy2	6	4141.3	4183.1	-2064.7	4129.3	6.9271	2	0.03132 *
accuracy3	6	4145.2	4187.0	-2066.6	4133.2	0.0000	0	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

LMMs

 Important!

The use of the `anova()` function is just for the convenience of having all the information on the same page!

```
anova(logtime1, logtime2, logtime3)
```

Data: data

Models:

```
logtime1: log(latency) ~ 0 + condition + (1 | stimuli) + (1 | respondent)
logtime2: log(latency) ~ 0 + condition + (0 + condition | stimuli) + (1 | respondent)
logtime3: log(latency) ~ 0 + condition + (1 | stimuli) + (0 + condition | respondent)
```

	npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
logtime1	5	6073.2	6108.0	-3031.6	6063.2			
logtime2	7	6061.4	6110.2	-3023.7	6047.4	15.743	2	0.0003815 ***
logtime3	7	5657.2	5705.9	-2821.6	5643.2	404.249	0	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Useful Models:

GLMMs

Model 2

θ_p

b_{WGBB} and b_{BGWB}

The IAT effect is mostly due to variations in the *stimuli functioning* between conditions, while the performance of the respondents seems unaltered

Results should be interpreted together!

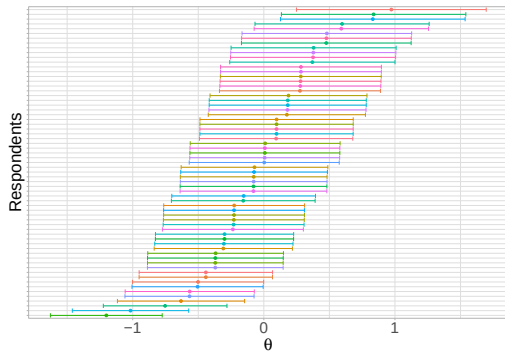
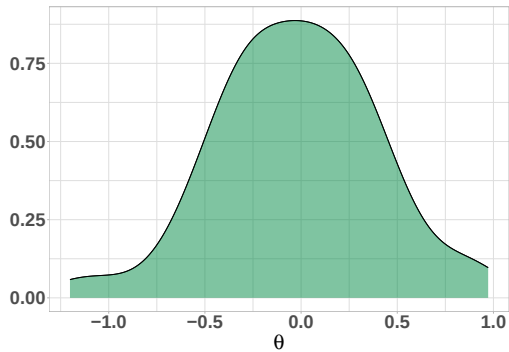
LMMs

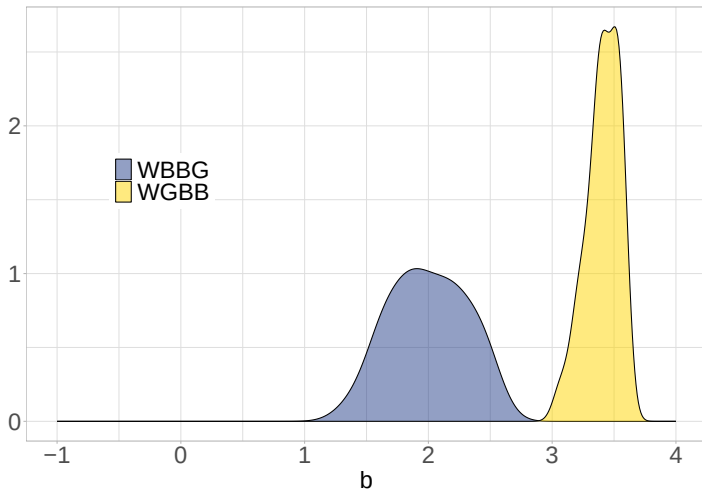
Model 3

τ_{WGBB} and τ_{BGWB}
 δ_s

The IAT effect is mostly due to variations in the *performance of the respondents* between conditions, while the functioning of the stimuli appears not affected

RASCH-LIKE ESTIMATES

 θ_p 

b_{WGBB} and b_{WGBB} 

FULLY-CROSSED STRUCTURES
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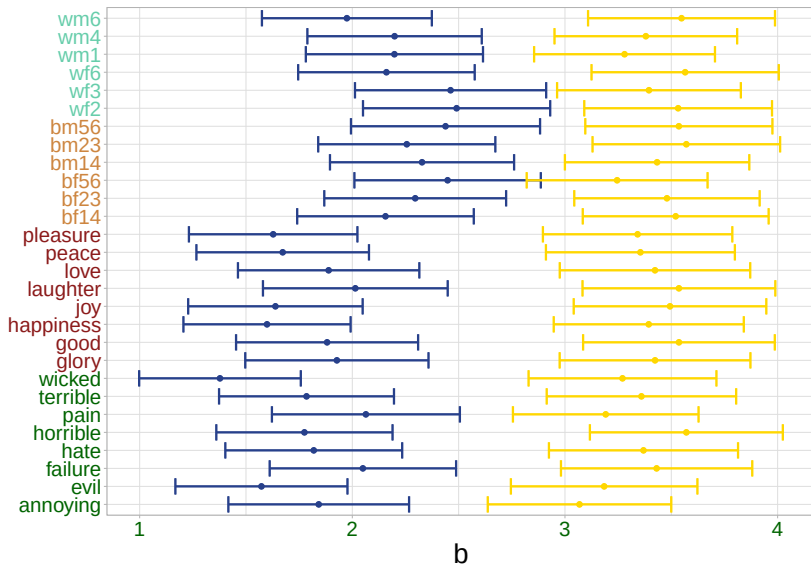
A CLASSIC OF PSYCHOMETRICS
○○○○○

RANDOM FACTORS AND EFFECTS
○○○○○○○

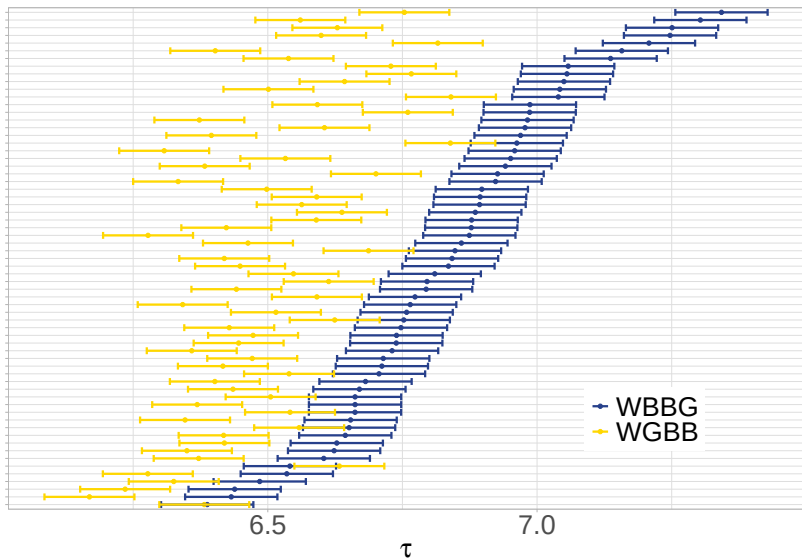
REAL DATA
○○○○○○○○○○○○○○●○○○

THE END
○○

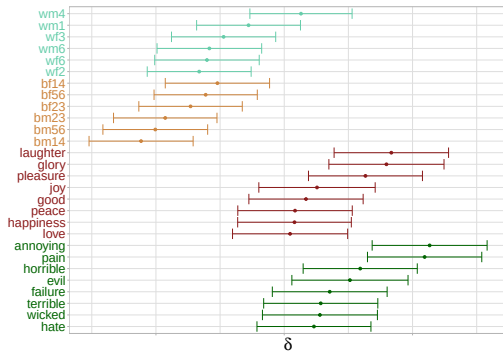
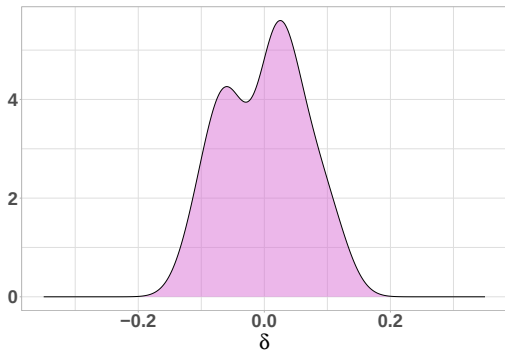
RASCH-LIKE ESTIMATES



LOG-NORMAL ESTIMATES



LOG-NORMAL ESTIMATES

 δ_s 

- The best model depends on the other models... sometimes useful, never right
- The sky is the limit... but do not over complicate things

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HOWEVER

- Time and accuracy are independent from one another, pretty bold assumption